Project SAPPHIRE: A Multi-Modal Framework for Perceptual Music Retrieval

Semantic and Acoustic Perceptual Holistic Integration REtrieval

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Abstract

This report details Project SAPPHIRE, an initiative to bridge the "perceptual gap" between the objective, computational analysis of music and the subjective, holistic nature of human music perception. Current Music Information Retrieval (MIR) systems are adept at analyzing physical properties but fail to capture perceived qualities like emotion, timbre, and groove. This failure is largely due to a reliance on flawed, rigid labels and a unimodal (acoustic-only) focus. Our vision is to model the *process* of human perception using a psychoacoustically-informed, multi-modal framework. This report outlines our methodology, which leverages Cross-Modal Contrastive Learning (CMCL), details the project execution through data pipeline construction, multi-modal feature extraction, and in-depth exploratory data analysis (EDA). We establish an empirical baseline (Jaccard Similarity: 0.0320) that quantifies the perceptual gap and validate our feature set through a successful mood classification task.

Contents

1 The Problem: The Perceptual Gap

Traditional Music Information Retrieval (MIR) systems operate on a fundamentally different level than human listeners. This creates a "perceptual gap," where machines can hear and analyze, but they do not intuitively *listen*.

1.1 The Disconnect Between Machine and Human

The core challenge lies in the disconnect between objective computational analysis and the subjective, nuanced nature of human music perception.

- **Machines Analyze:** Physical properties such as frequency (Hz), amplitude (dB), and raw waveforms.
- **Humans Perceive:** Subjective attributes such as pitch, loudness, emotion, timbre, and the elusive "groove" or "feel."

Current systems are experts at analysis, but novices at perception.

1.2 Why Traditional MIR Systems Fail

This gap is perpetuated by two foundational flaws in existing models:

1. Reliance on Flawed Labels Models are often trained on predefined genres (e.g., "Rock," "Jazz") and moods (e.g., "Happy," "Sad") that are rigid, subjective, and culturally dependent. Human annotators often disagree on these labels, creating an inconsistent and unreliable "ground truth." This imposes a hard performance ceiling on AI models that can only be as good as their flawed data.

2. Uni-Modal (Acoustic-Only) Focus Human listening is a holistic, multi-modal experience. We instinctively combine the sound with lyrics, assess the vocal performance, and understand the song's structure. Existing systems that focus only on acoustics ignore this critical context, leading to a shallow and incomplete understanding.

2 Our Vision and Methodology

2.1 A Perceptually-Driven Framework

Our vision is to model the **process of human perception**, not static labels. This is achieved through a framework built on three pillars:

- **Psychoacoustically-Informed Features:** Selecting features that correlate with human perception (e.g., spectral centroid for "brightness").
- **Holistic Multi-Modal Integration:** Combining acoustic, lyrical, rhythmic, and harmonic information.
- **Perceptually-Aligned Learning:** Using models that learn relationships and context, not just rigid classifications.

2.2 Core Methodology

Cross-Modal Contrastive Learning (CMCL) Instead of relying on flawed labels, our core methodology is built on CMCL. This model learns by comparing pairs of data (e.g., audio and its corresponding text). It learns to "pull" related audio and text snippets together into a shared, high-dimensional embedding space while "pushing" unrelated pairs apart. This allows the model to discover the emergent, perceptual links between sound and meaning on its own.

Large Language Models (LLMs) for Data Synthesis A significant challenge is the scarcity of rich, multi-modal music datasets. We leverage LLMs to overcome this by generating rich, context-aware descriptions for audio tracks, effectively synthesizing the high-quality text data needed for our CMCL model.

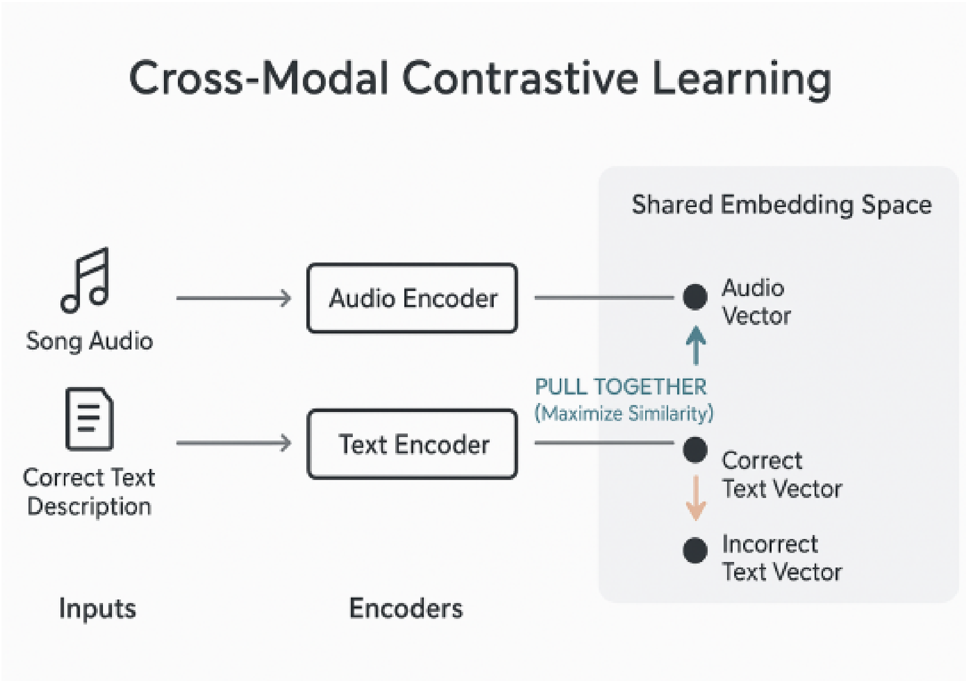

Figure 1: Proposed CMCL Architecture, which learns a shared embedding space for audio and lyrical text.

3 Project Execution and Data Analysis (Sprints 1–3)

3.1 Data Pipeline (Sprint 2)

We established a robust pipeline to process raw data into a structured corpus.

- **Dataset:** Zalo2022 Lyric Alignment dataset.
- **Processing:** Audio was resampled to 44.1 kHz and normalized to -23 LUFS (EBU R128) for perceptual consistency. Lyrics were standardized to UTF-8.
- **Final Corpus:** 264 high-quality, aligned audio-lyric tracks.

3.2 Phase 2: Multi-Modal Feature Extraction

We extracted a comprehensive set of features spanning all relevant perceptual domains, as detailed in Table 1.

3.3 Phase 3: Exploratory Data Analysis (EDA)

EDA revealed important biases and characteristics of our corpus. Univariate analysis showed distributions skewed towards brightness and moderate lyrical complexity.

Table 1: Overview of Extracted Multi-Modal Features

Domain	**Features**
Acoustic & Timbral	MFCCs, Spectral Centroid, Spectral Bandwidth, Spectral Rolloff, RMS Energy
Harmonic & Tonal	Chroma Features, Musical Key, Tonal Complexity
Rhythmic	Tempo (BPM), Rhythm Regularity, Syncopation Index
Lyrical	Type-Token Ratio (TTR), Readability Scores, Sentiment (VADER)
Audio Quality	Signal-to-Noise Ratio (SNR), Dynamic Range

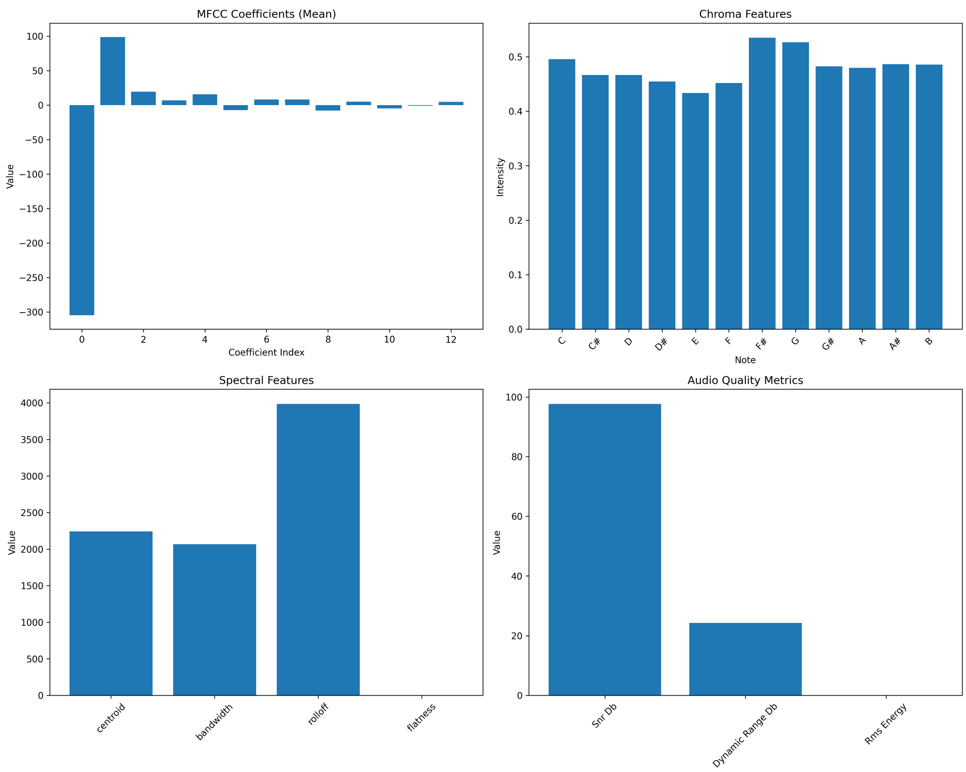

Figure 2: Analysis of feature metrics across the dataset.

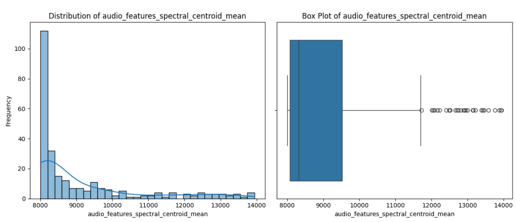

Figure 3: Distribution of key acoustic features.

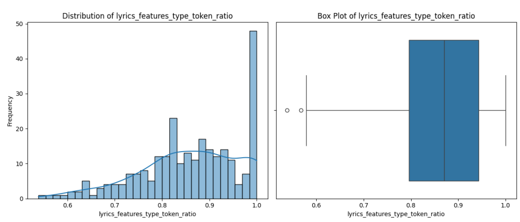

Figure 4: Distribution of key lyrical features.

Cross-modal correlation analysis (Figure 5) revealed that in their raw form, sound and meaning are only weakly correlated. This finding motivates the need for a sophisticated model (like CMCL) to find the non-linear, latent relationships.

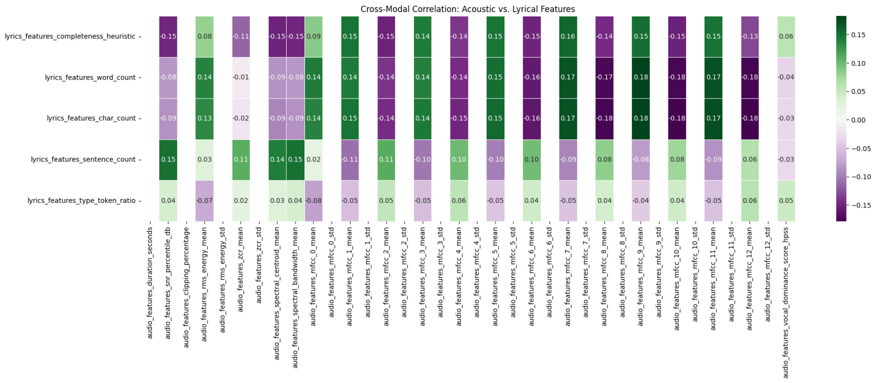

Figure 5: Correlation analysis showing weak linear relationships between raw acoustic and lyrical features.

3.4 Uncovering the Latent Sonic Space (Sprint 3)

We used Principal Component Analysis (PCA) to distill the complex, high-dimensional acoustic features into their most essential axes of variation. This revealed a striking result: the first two Principal Components (PCs) alone capture **80.83%** of the total acoustic variance.

- **PC1:** Represents Timbral Complexity & Density (Bright/Rich vs. Dark/Simple).
- **PC2:** Represents Rhythmic Energy & Percussiveness (High-energy/Noisy vs. Smooth/Tonal).

This confirms that the complex "sound" of a song can be meaningfully compressed into a much lower-dimensional space, a foundational concept for our model.

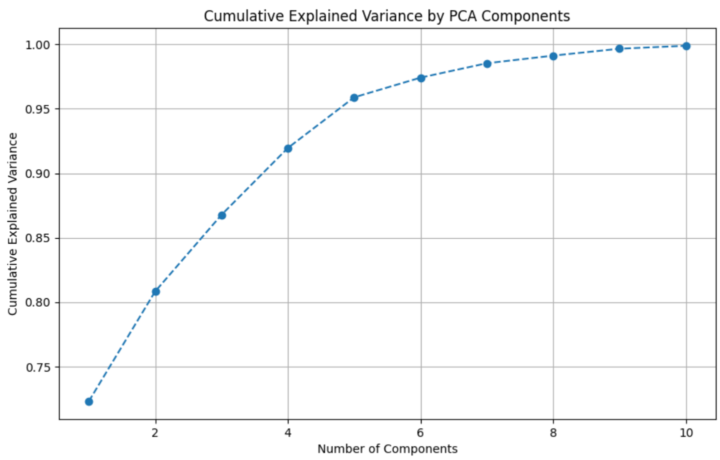

Figure 6: PCA cumulative explained variance, with the first two components capturing 80.83%.

Furthermore, clustering tracks based on these acoustic PCs revealed emerging semantic patterns in their lyrics, suggesting that sonically similar groups share latent meaning (Figure 7).

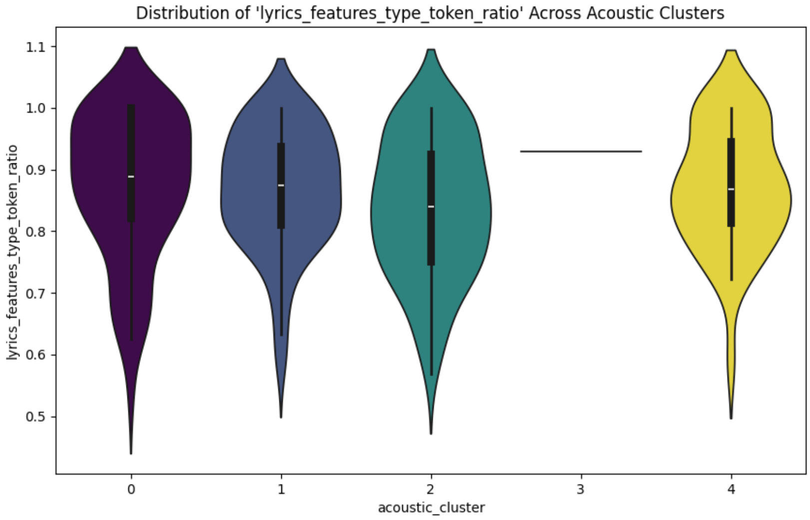

Figure 7: Clustering analysis showing that sonic groups (formed by PCA) exhibit distinct lyrical feature distributions.

4 Empirical Baseline: Quantifying the Gap

To validate our project's core motivation, we quantified the perceptual gap. We created two "neighbor" sets for each song:

1. **Acoustic Neighbors:** The 10-NN using Euclidean distance in the acoustic feature space.
2. **Semantic Neighbors:** The 10-NN using cosine similarity in the lyrical (embedding) space.

We then measured the overlap between these two sets using the Jaccard similarity coefficient.

The result was an average Jaccard similarity of **0.0320**. This extremely low score provides a stark, empirical validation of our central hypothesis: **acoustic similarity does not imply semantic similarity** in existing feature spaces.

5 Application: Mood Classification

Having established our feature set, we tested its practical utility on a real-world MIR task: mood classification.

5.1 Task, Dataset, and Model

- **The Task:** Train a model to predict a song's mood based on our extracted multi-modal features.
- **The Dataset:** A MIREX-like Mood Dataset with expert annotations for 5 distinct mood clusters (e.g., "Passionate/Rousing," "Bittersweet/Wistful").
- **The Model:** We trained a Random Forest classifier, chosen for its robustness and high interpretability.

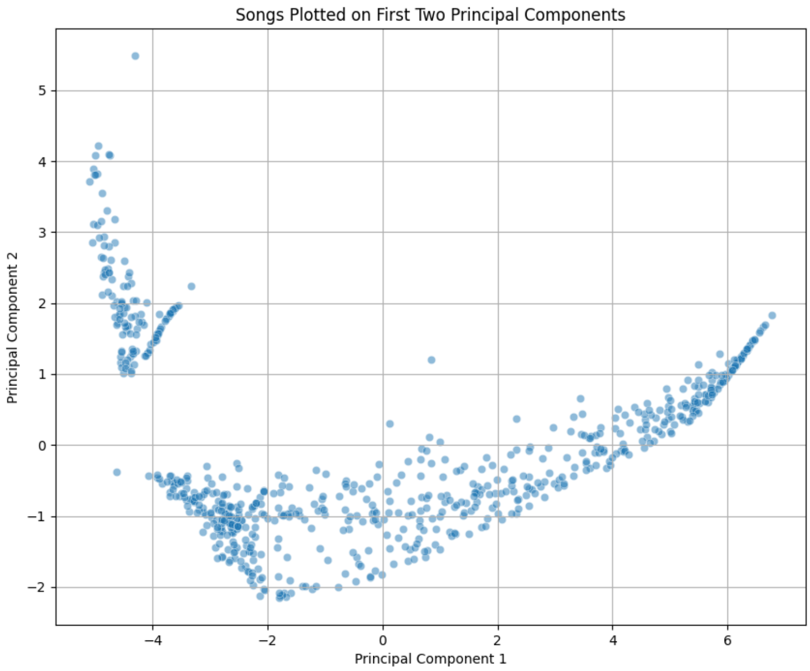

Figure 8: Visualization of mood clusters in the latent sonic space (PC1 vs. PC2).

5.2 Key Insights from Mood Classification

The feature importance analysis from the trained model revealed what truly drives mood perception in the data.

- **Finding 1: Multi-Modality is Crucial.** The most predictive features were a mix of acoustic, lyrical, and rhythmic domains. An acoustic-only model would miss critical information.
- **Finding 2: Energy and Emotion Lead. RMS Energy** (loudness) and **Spectral Centroid** (brightness) were top acoustic predictors. Lyrical **Sentiment** (VADER score) was the most important semantic feature.
- **Finding 3: Rhythm and Harmony Provide Nuance.** Features like **Tempo** and **Tonal Complexity** were also significant, confirming their contribution to the overall emotional character.

5.3 Final Psychoacoustically-Informed Feature Set

Based on this analysis, we finalized our feature set, focusing on features with high perceptual correlation.

Timbre & Texture

- **MFCCs:** Captures the timbral "fingerprint" or color of the sound.
- **Spectral Centroid:** Correlates with the perceived "brightness."
- **RMS Energy:** Represents the perceived loudness and intensity.

Rhythm & Structure

- **Tempo (BPM):** The fundamental pace or "heartbeat" of the music.
- **Rhythm Regularity:** Measures the steadiness of the beat, influencing "groove."
- **Chroma Features:** Represents the tonal content (C, C#, D...), key to harmony.

Semantic Content

- **Sentiment Score:** Quantifies the emotional tone of the lyrics.
- **Readability Score:** Indicates lyrical complexity (simple vs. abstract).
- **Semantic Embeddings:** A vector representing the core themes and meaning.

6 Current Work and Future Roadmap

6.1 Current Progress: Dataset Expansion

Our recent work has focused on dataset expansion to broaden the scope and robustness of our model. Key additions include:

- **JAM-ALT (Jamendo Alternative):** A multi-modal dataset (https://huggingface.co/datasets/jamendolyrics/jam-alt) linking lyrics and audio, ideal for contrastive learning.
- **CSD (Cross-Modal-Similarity Dataset):** Contains audio features and rich semantic annotations (https://zenodo.org/records/4785016), crucial for bridging the semantic-acoustic gap.
- **MIREX MOOD:** An expert-annotated dataset for mood (https://mir.dei.uc.pt/downloads.html), serving as a key benchmark for our classification tasks.

6.2 Project Timeline

The project remains on track with our original 12-week plan, as shown in Figure 9. We are currently finalizing our features and moving into the core modeling phase.

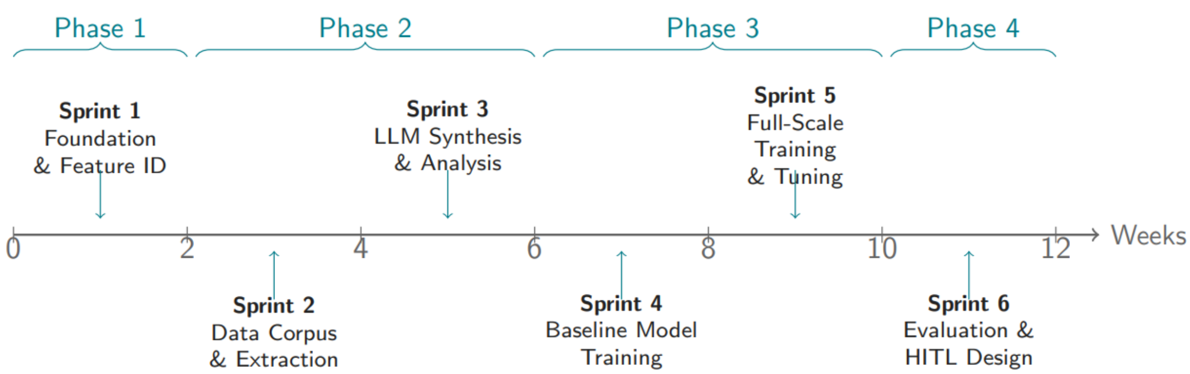

Figure 9: Project Timeline and Key Milestones.

6.3 Conclusion and Next Steps

Achievements to Date

- Established a robust, multi-modal data pipeline.
- Conducted in-depth EDA and latent space analysis.
- Quantified the perceptual gap with an empirical baseline (Jaccard: **0.0320**).
- Validated our multi-modal feature set via a mood classification task.

Path Forward

- Finalize the optimal feature set based on our cross-dataset analysis.
- Implement the full Cross-Modal Contrastive Learning (CMCL) model.
- Evaluate the model's performance against our baseline and other MIR systems.
- Develop the final product: a perceptually-aligned music retrieval system.